5-MEMBER AGILE SCRUM SPECIFICATION • DEPARTMENT OF DATA SCIENCE (PUCIT)

AutoCare AI: 5-Member Jira Sprint Backlog & WBS

Formal Distribution across 5 Students (32 SP Each) | 4 Sprints | 160 Story Points Total

Academic Scope:	PUCIT BS Data Science Final Year Project / Big Data Systems
Jira Key & Repo:	Key: **AUT** | Remote: github.com/marketbul/autocare-ai.git
5 Team Members:	Abdullah (035), Talha (016), Danial (014), Ahmad (033), M. Abdullah (032)
Velocity & Balance:	160 SP Total | Exactly 32 SP per member (8 SP per sprint) | 100% Balanced

1. 5-Member Engineering Roles & Core Ownership

Member	Roll No	Primary Role	Assigned Epics	Total SP
Abdullah (Lead)	BSDSF23A035	System Architect & RAG	Consistent Hashing, Causal Transformer LM	**32 SP**
Talha Ali Bhatti	BSDSF23A016	Storage & Ingestion	MinIO S3, CAS Deduplication, Parquet Lakehouse	**32 SP**
Danial Hameed	BSDSF23A014	Data & Multilingual	Dataset Ingestion, Bloom Filter, Bilingual Lexicon	**32 SP**
Ahmad Zahid	BSDSF23A033	ML Triage & Calib	Cost-Sensitive Triage, Platt Scaling, NLI Verifier	**32 SP**
Muhammad Abdullah	BSDSF23A032	Frontend & Analytics	DuckDB OLAP, Modern Web Portal UI, PKR Pricing	**32 SP**

2. 4-Week Sprint Schedule & Velocity Allocation (160 SP)

Sprint	Timeline	Abdullah	Talha	Danial	Ahmad	M. Abd	Sprint SP
Sprint 1	Days 1–7	8 SP	8 SP	8 SP	8 SP	8 SP	**40 SP**
Sprint 2	Days 8–14	8 SP	8 SP	8 SP	8 SP	8 SP	**40 SP**
Sprint 3	Days 15–21	8 SP	8 SP	8 SP	8 SP	8 SP	**40 SP**
Sprint 4	Days 22–28	8 SP	8 SP	8 SP	8 SP	8 SP	**40 SP**
Total	**4 Weeks**	**32 SP**	**32 SP**	**32 SP**	**32 SP**	**32 SP**	**160 SP**

3. Complete 5-Member Jira Ticket Catalog (40 Tickets, AUT-101 – AUT-143)

Key	Sprint	Summary & Git Commit Tag	SP	Assignee	Status
AUT-104	Sprint 1	**Consistent Hash Ring with 160 Virtual Nodes per Physical Node** [AUT-104] feat(algorithms): consistent hash ring with 160 virtual nodes and Murmur3	**8**	Abdullah	**Done**
AUT-101	Sprint 1	**MinIO S3 Distributed Multi-Bucket Cluster Architecture** [AUT-101] feat(storage): provision distributed MinIO S3 cluster and bucket policies	**5**	Talha	**Done**
AUT-102	Sprint 1	**SHA-256 Content-Addressable Storage (CAS) Deduplication Engine** [AUT-102] feat(storage): implement SHA-256 content-addressable storage deduplication	**3**	Talha	**Done**
AUT-115	Sprint 1	**Automotive Dataset Ingestion & Unified Entity Schema Standardization** [AUT-115] feat(data): clean, preprocess and standardize 111 automotive dataset files	**5**	Danial	**Done**
AUT-103	Sprint 1	**Structure-Aware Technical Manual Chunking & Metadata Extraction** [AUT-103] feat(ingest): structure-aware PDF chunking with OEM metadata headers	**3**	Danial	**Done**
AUT-138	Sprint 1	**Supervised Fault Category & Severity Triage Classifier** [AUT-138] feat(ml): train supervised fault category and severity triage classifier	**5**	Ahmad	**Done**
AUT-139	Sprint 1	**Cost-Sensitive Loss Matrix (10x Safety Penalty on Critical Errors)** [AUT-139] feat(ml): cost-sensitive loss weighting with 10x safety penalty matrix	**3**	Ahmad	**Done**
AUT-114	Sprint 1	**DuckDB In-Memory OLAP Query Engine Integration for Parquet** [AUT-114] feat(analytics): embed DuckDB OLAP engine for zero-copy parquet querying	**5**	Muhammad	**Done**
AUT-116	Sprint 1	**PySpark Distributed DataFrame Windowed Transformations Pipeline** [AUT-116] feat(analytics): PySpark distributed windowed transformations pipeline	**3**	Muhammad	**Done**
AUT-105	Sprint 2	**Consistent Hash Node Dynamic Rebalancing & Binary Ring Routing** [AUT-105] feat(algorithms): bisect ring lookup and dynamic cluster rebalancing logic	**5**	Abdullah	**Done**
AUT-128	Sprint 2	**FastAPI Asynchronous Gateway & Microservice Routing Architecture** [AUT-128] feat(api): asynchronous FastAPI gateway with JWT auth and OpenAPI specs	**3**	Abdullah	**Done**
AUT-108	Sprint 2	**OBD-II CAN-Bus High-Velocity Telemetry Stream Generator** [AUT-108] feat(ingest): high-velocity OBD-II CAN-bus synthetic telemetry generator	**5**	Talha	**Done**
AUT-110	Sprint 2	**Snappy Columnar Parquet Lakehouse Partitioning (Make/Model/Year/Region)** [AUT-110] feat(lakehouse): snappy columnar parquet writer with hierarchical partitioning	**3**	Talha	**Done**
AUT-106	Sprint 2	**Counting Bloom Filter for O(1) Set Membership & Redundant I/O Gate** [AUT-106] feat(algorithms): counting bloom filter with k-hash double-hashing gate	**5**	Danial	**Done**
AUT-107	Sprint 2	**Bloom Filter Bit-Array Serialization & Redis State Sync** [AUT-107] feat(algorithms): bloom filter binary serialization and snapshot restore	**3**	Danial	**Done**
AUT-140	Sprint 2	**Platt Scaling & Temperature Confidence Calibration (ECE Optimization)** [AUT-140] feat(calibration): implement Platt scaling and Expected Calibration Error (ECE) minimizer	**5**	Ahmad	**Done**
AUT-141	Sprint 2	**Grouped K-Fold Cross-Validation Protocol across Vehicle Models** [AUT-141] test(ml): implement Grouped K-Fold validation across vehicle chassis groups	**3**	Ahmad	**Done**
AUT-117	Sprint 2	**Fleet Diagnostic Trouble Code (DTC) Anomaly Detection Model** [AUT-117] feat(ml): unsupervised Isolation Forest anomaly detection for OBD-II telemetry	**5**	Muhammad	**Done**
AUT-118	Sprint 2	**Regional Component Degradation & Pakistani Climate Risk Multipliers** [AUT-118] feat(analytics): regional climate risk multipliers and component wear equations	**3**	Muhammad	**Done**
AUT-121	Sprint 3	**Custom Subword Byte-Pair Encoding (BPE) Tokenizer from Scratch** [AUT-121] feat(rag-scratch): subword BPE tokenizer with 4000 vocab and special tokens	**5**	Abdullah	**Done**
AUT-123	Sprint 3	**4-Layer Causal Transformer Decoder with Multi-Head Self-Attention** [AUT-123] feat(rag-scratch): 4-layer causal transformer decoder with multi-head attention	**3**	Abdullah	**Done**

AUT-109	Sprint 3	**In-Memory Sliding-Window Batch Buffer & Ingestion Sink** [AUT-109] feat(ingest): sliding-window telemetry buffer and micro-batch flusher	**5**	Talha	**Done**
AUT-111	Sprint 3	**Automated Parquet Compaction Job & Small File Merging Engine** [AUT-111] feat(lakehouse): parquet small-file compaction and catalog vacuum engine	**3**	Talha	**Done**
AUT-136	Sprint 3	**Multilingual Automotive Lexicon (Urdu & Roman Urdu to Standard DTCs)** [AUT-136] feat(nlp): bilingual Urdu and Roman Urdu automotive mechanical term translator	**5**	Danial	**Done**
AUT-122	Sprint 3	**Dense L2-Normalized Vector Embedding Neural Network (D=128)** [AUT-122] feat(rag-scratch): dense L2-normalized vector embedding neural network (D=128)	**3**	Danial	**Done**
AUT-132	Sprint 3	**Workshop Repair Invoice Verifier & Non-Accusatory NLI Engine** [AUT-132] feat(verifier): non-accusatory workshop invoice verifier with NLI evidence cross-check	**5**	Ahmad	**Done**
AUT-142	Sprint 3	**Repair Fraud & Unnecessary Parts Replacement Detection Heuristics** [AUT-142] feat(verifier): predatory repair detection heuristics and consumer protection rules	**3**	Ahmad	**Done**
AUT-119	Sprint 3	**DuckDB OLAP Benchmark vs Traditional Relational SQL (15x Speedup)** [AUT-119] test(benchmark): comparative query latency benchmark of DuckDB vs SQLite	**5**	Muhammad	**Done**
AUT-120	Sprint 3	**Pakistani Fair-Price Market Estimator (PKR Parts & Mechanic Labor)** [AUT-120] feat(pricing): localized PKR parts and mechanic labor fair-price estimator	**3**	Muhammad	**Done**
AUT-124	Sprint 4	**Autoregressive LM Training Pipeline with AdamW & Perplexity Optimization** [AUT-124] feat(rag-scratch): autoregressive LM training pipeline with AdamW and Perplexity tracking	**5**	Abdullah	**Done**
AUT-126	Sprint 4	**Multilingual Grounded RAG Generation & Anti-Hallucination Pipeline** [AUT-126] feat(rag-scratch): end-to-end grounded RAG generation with selective abstention	**3**	Abdullah	**Done**
AUT-113	Sprint 4	**Storage IOPS & Ingestion Latency Percentile Benchmark Profiler** [AUT-113] perf(storage): automated IOPS and ingestion latency benchmarking suite	**5**	Talha	**Done**
AUT-112	Sprint 4	**Lakehouse Schema Evolution & Corrupt Record Dead-Letter Sink** [AUT-112] feat(lakehouse): schema evolution handler and dead-letter quarantine sink	**3**	Talha	**Done**
AUT-125	Sprint 4	**Hybrid Vector Store (Dense Cosine Similarity + BM25 Lexical Ranking)** [AUT-125] feat(rag-scratch): hybrid dense vector + BM25 lexical retrieval engine	**5**	Danial	**Done**
AUT-137	Sprint 4	**Reciprocal Rank Fusion (RRF) & Context Re-Ranking Layer** [AUT-137] feat(retrieval): Reciprocal Rank Fusion (RRF) multi-list reranking layer	**3**	Danial	**Done**
AUT-143	Sprint 4	**Selective Abstention Gate & Uncertainty Threshold Governance** [AUT-143] feat(safety): selective abstention gate and confidence threshold governance	**5**	Ahmad	**Done**
AUT-129	Sprint 4	**End-to-End System Safety Validation & Pytest Test Suite** [AUT-129] test(safety): complete pytest test suite for triage safety and abstention gates	**3**	Ahmad	**Done**
AUT-127	Sprint 4	**Modern Automotive Diagnostic Web Portal Redesign (React 18 + Tailwind)** [AUT-127] feat(frontend): modern automotive engineering web portal with diagnostic terminal	**5**	Muhammad	**Done**
AUT-130	Sprint 4	**Final Academic Project Initiation Document (PID) & Defense Presentation** [AUT-130] docs(academic): generate professional 4-week Big Data PID in PDF and Word	**3**	Muhammad	**Done**